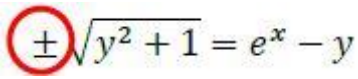


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies the $\pm$ sign (or just the negative) as the error. May be seen in any of the last five lines. May be indicated within Matthew's solution or described in words. Ignore other 'errors' identified. Condone identifying any of the last five lines as containing the error. PI	AO2.3	B1	
	Gives a correct reason, referring to either e^x (or e^y), or the operand of a logarithm, being positive. Do not award if more than one error identified.	AO2.4	E1	It is an error because $y - \sqrt{y^2 + 1} < 0$ and $e^x > 0$ so there is a contradiction.
(b)	States the correct solution of the equation. Accept 10.0 [1787493] or $\frac{1}{2}(e^3 - e^{-3})$ ISW	AO1.1b	B1	$\operatorname{arsinh} x = 3$ $x = \sinh 3$
				Total 3 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses definitions of $\sinh x$ and $\cosh x$ to obtain expression for $\tanh x$	AO1.2	B1	$\sinh x = \frac{e^x - e^{-x}}{2}$
	Multiplies by e^x	AO1.1a	M1	$\cosh = \frac{e^x + e^{-x}}{2}$
	Obtains e^{2x}	AO1.1b	A1	$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
	Completes a fully correct argument by demonstrating result by taking logs This mark is available only if all	AO2.1	R1	Multiplying numerator and denominator by e^x $t = \frac{e^{2x} - 1}{e^{2x} + 1}$

previous marks have been awarded			$te^{2x} + t = e^{2x} - 1$ [or multiplies by e^x in $te^x + te^{-x} = e^x - e^{-x}$ $1 + t = e^{2x}(1 - t)$ $e^{2x} = \frac{1+t}{1-t}$ $2x = \ln \frac{1+t}{1-t}$ hence $x = \frac{1}{2} \ln \frac{1+t}{1-t}$
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(b)	Expresses cosh 3x and cosh x in exponential form Seen anywhere in solution	AO1.2	B1	$\left(\frac{e^x + e^{-x}}{2} \right)^3 =$ To be proven: $\frac{1}{4} \left(\frac{e^{3x} + e^{-3x}}{2} \right) + \frac{3}{4} \left(\frac{e^x + e^{-x}}{2} \right)$
(i)	Expands LHS FT 'their' LHS provided first M1 awarded Allow one slip	AO1.1a	M1	$\text{LHS} \left(\frac{e^x + e^{-x}}{2} \right)^3 =$ $\frac{1}{8} (e^{3x} + 3e^{2x} \cdot e^{-x} + 3e^x \cdot e^{-2x} + e^{-3x}) =$
	Simplifies and collects terms FT 'their' expression ©2025 Exam Papers Practice. All rights reserved. Allow one slip	AO1.1a	M1	$\frac{1}{4} \left(\frac{e^{3x} + e^{-3x}}{2} \right) + \frac{3}{4} \left(\frac{e^x + e^{-x}}{2} \right) =$ $\frac{1}{4} \cosh 3x + \frac{3}{4} \cosh x = \text{RHS}$
	Completes fully correct proof to reach the required result This mark is available only if all previous marks have been awarded	AO2.1	R1	From the definition of cosh x
(ii)	Substitutes for cosh 3x in equation from part (a) Allow one slip	AO3.1a	M1	$(\cosh x)^3 = \frac{1}{4} \times 13 \cosh x + \frac{3}{4} \cosh x$ $(\cosh x)^3 - 4 \cosh x = 0$
	Obtains equation in cosh x and solves it	AO1.1a	M1	

Allow one slip			$\cosh x[(\cosh x)^2 - 4] = 0$
Eliminates 0 and -2 with reason	AO2.4	E1	Solutions are $\cosh x = 0, -2, 2$
States correct solution in exact log form	AO1.1b	A1	solutions 0 and -2 are not possible since range of $\cosh x \geq 1$ $\cosh x = 2 \Rightarrow x = \ln(2 + \sqrt{3})$
Total 12 marks			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Commences proof by considering one side of the identity only: if considering LHS combines terms as a single fraction with a common denominator. If considering RHS writes $\coth \theta$ as a fraction and introduces factor of $1 + \cosh \theta$ to both numerator and denominator Note alternative valid approaches include commencing proof by considering LHS - RHS or LHS divided by RHS	AO2.1	R1	$\frac{\sinh \theta}{1 + \cosh \theta} + \frac{1 + \cosh \theta}{\sinh \theta}$ $= \frac{\sinh^2 \theta + 1 + \cosh^2 \theta + 2 \cosh \theta}{(1 + \cosh \theta) \sinh \theta}$ $= \frac{\cosh^2 \theta + \cosh^2 \theta + 2 \cosh \theta}{(1 + \cosh \theta) \sinh \theta}$ $\because 1 + \sinh^2 \theta \equiv \cosh^2 \theta$ $= \frac{2 \cosh \theta (1 + \cosh \theta)}{(1 + \cosh \theta) \sinh \theta}$ $= \frac{2 \cosh \theta}{\sinh \theta}$
	Explicitly states identity $\cosh^2 \theta - \sinh^2 \theta \equiv 1$ and uses it to eliminate (or introduce) $\sinh^2 \theta$	AO2.4	R1	$\equiv 2 \coth \theta$ AG
	Factorises numerator and cancels correctly for 'their' fraction (if considering RHS rearranges 'their' numerator correctly into two factorised expressions)	AO1.1b	B1F	
	Completes rigorous proof to obtain result AG Only award if they have a completely correct argument, which is clear and contains no slips.	AO2.1	R1	

(b)	Uses result from part (a) to deduce that $\tanh \theta = \frac{1}{2}$	AO2.2a	R1	$2\coth \theta = 4$ $\tanh \theta = \frac{1}{2}$
	Uses natural log form and substitutes correct value to obtain correct exact form	AO1.1b	A1	$\theta = \tanh^{-1} \frac{1}{2} = \frac{1}{2} \ln 3$
Total 6 marks				

Q4.

(a) $x = \tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$

$$xe^y + xe^{-y} = e^y - e^{-y}$$

$$\text{or } xe^{2y} + x = e^{2y} - 1$$

M1

$$\Rightarrow (x+1)e^{-y} = e^y(1-x)$$

$$\Rightarrow (x+1) = e^{2y}(1-x)$$

A1

$$e^{2y} = \frac{1+x}{1-x} \Rightarrow y = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

AG

A1cso

(b) $y = \frac{1}{2} \ln(1+x) - \frac{1}{2} \ln(1-x)$

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$$\frac{dy}{dx} = \frac{1}{2(1+x)} + \frac{1}{2(1-x)}$$

A1

$$= \frac{1-x+1+x}{2(1+x)(1-x)} = \frac{2}{2(1-x^2)} = \frac{1}{1-x^2}$$

AG

A1cso

Alternative 1

$$\frac{dy}{dx} = \frac{1}{2} \times \frac{(1-x)}{(1+x)} \times \frac{d}{dx} \left(\frac{1+x}{1-x} \right) \quad \text{M1}$$

$$\frac{dy}{dx} = \frac{1}{2} \times \frac{(1-x)}{(1+x)} \times \frac{(1-x) + (1+x)}{(1-x)^2} \quad \text{A1}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1-x^2} \quad \text{A1 cso}$$

(c) $\int 4x \tanh^{-1} x \, dx = 4x \tanh^{-1} x - \int \frac{4x}{1-x^2} \, dx$

M1

$$4x \tanh^{-1} x + 2 \ln(1 - x^2)$$

A1

$$\tanh^{-1} \frac{1}{2} = \frac{1}{2} \ln 3$$

must simplify logarithm to $\ln 3$

B1

Value of integral = $\ln 3 + 2 \ln \frac{3}{4}$

any correct form

A1

$$\ln \left(\frac{3^3}{2^4} \right)$$

all working must be correct

A1cso

5

[11]

Q5.

- (a) Sketch, approximately correct shape

B1

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Asymptotes at $y = \pm 1$

*B0 if curve touches asymptotes
lines could be used for asymptotes
be strict with sketch*

B1

2

(b) Use of $u = \frac{\sinh x}{\cosh x}$

M1

$$= \frac{e^x - e^{-x}}{e^x + e^{-x}} \text{ or } \frac{e^{2x} - 1}{e^{2x} + 1}$$

A1

$$u(e^x + e^{-x}) = e^x - e^{-x}$$

M1 for multiplying up

M1

$$e^{-x}(1 + u) = e^x(1 - u)$$

A1 for factorizing out e's or M1 for attempt at 1 + u and 1 - u in terms of e^x

A1

$$e^{2x} = \frac{1+u}{1-u}$$

m1

$$x = \frac{1}{2} \ln \left(\frac{1+u}{1-u} \right)$$

AG

A1

6

- (c) (i) Use of $\tanh^2 x = 1 - \operatorname{sech}^2 x$

M1

Printed answer

A1

2

- (ii) $(3 \tanh x - 1)(\tanh x - 2) = 0$

Attempt to factorise

M1

$$\tanh x \neq 2$$

Accept $\tanh x \neq 2$ written down but not ignored or just crossed out

E1

$$\tanh x = \frac{1}{3}$$

A1

$$x = \frac{1}{2} \ln 2$$

M1

ft

A1F

5

[15]

Q6.

(a) $5\left(\frac{e^x - e^{-x}}{2}\right) + \left(\frac{e^x + e^{-x}}{2}\right)$

M0 if no 2s in denominator

M1

$$= 3e^x - 2e^{-x}$$

A1

2

(b) $3e^x - 2e^{-x} + 5 = 0$
 $3e^{2x} + 5e^x - 2 = 0$

ft if 2s missing in (a)

M1

$$(3e^x - 1)(e^x + 2) = 0$$

A1F

$$e^x \neq -2$$

*any **indication** of rejection*

E1

$$e^x = \frac{1}{3} \quad x = \ln \frac{1}{3}$$

provided quadratic factorises into real factors

A1F

4

[6]

Q7.

(a) Use of $\cosh^2 x = 1 + \sinh^2 x$

Must be correct for M1

M1

$$4 \sinh^2 x - 7 \sinh x + 3 = 0$$

A1

$$(4 \sinh x - 3)(\sinh x - 1) = 0$$

Provided quadratic factorizes

A1 ✓

$$\sinh x = \frac{3}{4} \text{ or } 1$$

A1 ✓

4

(b) Use of formula for $\sinh^{-1}$

M1

$$x = \ln 2 \text{ or } \ln(1 + \sqrt{2})$$

A1 ✓

A1 ✓

3

[7]

Q8.

(a) (i)
$$\frac{e^k + e^{-k}}{2} - \frac{3(e^k - e^{-k})}{2} = -1$$

Allow if 2's are missing or if cosh x and sinh x interchanged

M1

$$-2e^k + 4e^{-k} = -2$$

A1

$$e^{2k} - e^k - 2 = 0$$

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A1

3

(ii) $(e^k + 1)(e^k - 2) = 0$

M1

$$e^k \neq -1$$

*Must state something to earn E1.
Do not accept ignoring or crossing out.*

E1

$$e^k = 2$$

A1

$$k = \ln 2$$

A1F

4

- (b) (i) $\cosh x = 3\sinh x$ or in terms of e^x

M1

$$\tanh x = \frac{1}{3} \text{ or } 2e^x = 4e^{-x}$$

A1

$$x = \frac{1}{2} \ln \left(\frac{1 + \frac{1}{3}}{1 - \frac{1}{3}} \right) \text{ or } e^{2x} = 2$$

A1F

$$x = \frac{1}{2} \ln 2$$

CAO

A1

4

(ii) $\frac{dy}{dx} = \sinh x - 3\cosh x$ or $-e^x - 2e^{-x}$

M1

$$= 0 \text{ when } \tanh x = 3 \text{ or } e^{2x} = -2$$

A1

Correct reason
Must give a reason

E1

3

$$\frac{d^2y}{dx^2} = y = 0 \text{ at } \left(\frac{1}{2} \ln 2, 0 \right)$$

- (iii) ie one point

B1F

1

[15]

Examiner reports

Q1.

Many students failed to identify the error in the algebraic steps given in part (a). Of those who realised that $\pm$ was not acceptable, only half could explain why. A significant number of students suggested that completing the square was an invalid approach. Many others questioned the swapping of the variables x and y , with some explaining that variables could not be swapped. Some students decided that the variables had been swapped at the wrong point in the solution.

When answering part (b), many students failed to notice the connection with part (a). Of those who did spot the link, some decided that x was equal to $\sinh^{-1} 3$ instead of $\sinh 3$. However, a number of students were able to solve the equation by algebraic means, and the correct answer sometimes followed several lines of skilful algebra.

Q4.

- (a) Most students used $x = \tanh y$ and the given exponential form to obtain an expression in terms of x for e^{2y} and hence the printed result. Quite a few students thought that $\tanh^{-1} y = \frac{\sinh^{-1} y}{\cosh^{-1} y}$ and made no progress. Those students who started with $\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ were also rarely successful.
- (b) The majority of students used the chain rule to differentiate the result from part (a). Those with greater insight used the law of logarithms to obtain the difference of two logarithms and the new form was then much easier to differentiate. Poor algebraic manipulation meant that some were forced to fudge their working to obtain the printed answer.
- (c) This part of the question was the best source of marks for most students, though some were unable to integrate $\frac{x}{1-x^2}$ correctly. Those whose calculus was sound had no trouble expressing the final answer in the given form.

Q5.

Sketches were poor in part (a). Sometimes asymptotes were not drawn and even when they were sketches crossed or mingled with their asymptotes. It was not uncommon to see $\frac{\pi}{2}$ or π on candidates' diagrams showing some confusion with the graph of $y = \tan x$.

In part (b), provided that candidates knew what to do when they reached $u = \frac{e^{2x} - 1}{e^{2x} + 1}$, they almost always went on to complete this part correctly, but a substantial number of solutions petered out at this point.

Part (c) was well done apart from the rejection of $\tanh x = 2$ where lack of adequate reasoning for its rejection was often apparent.

Q6.

Most candidates scored full marks for this question. The only significant error was in the use of incorrect formulae for $\sinh x$ and $\cosh x$. These formulae were sometimes quoted without the halves, ie $\sinh x$ was written as $e^x - e^{-x}$ and $\cosh x$ as $e^x + e^{-x}$. In part (b), a small number of candidates thought that $\ln(3e^{2x} + 5e^x - 2)$ was equal to $\ln 3e^{2x} + \ln 5e^x - \ln 2$.

Q7.

Apart from a few candidates who factorised the quadratic in $\sinh x$ incorrectly, most candidates worked part (a) correctly. However, many candidates spent more time on part (b) than was necessary. They expressed $\sinh x$ in exponential form and solved the ensuing quadratic equations rather than quote the formula for $\sinh^{-1} x$ given in the formulae booklet which they were entitled to do. This method also led to superfluous incorrect solutions which candidates needed to reject.

Q8.

Again, this question proved to be a good source of marks for many candidates. Some candidates in part (a)(i) mixed the exponential forms for $\cosh x$ and $\sinh x$, whilst others, having expressed $\cosh x$ and $\sinh x$ in exponential form correctly and having arrived at $-e^k + 2e^{-k} = -1$ were unable to take the final step which led to the printed result. There was, also, not always a very convincing reason for the rejection of $e^k = -\ln$ in this part of the question. A very common error in part (a)(ii) was to write $2k - k - \ln 2 = 0$ after the printed answer, leading to the correct answer by totally incorrect mathematics. Part (b)(i) was usually answered well, as was part (b)(ii), apart from those candidates who thought that the derivatives of hyperbolic functions followed the pattern of the derivatives of trigonometrical functions and so incurred sign errors. The explanation for the rejection of solutions to $e^{2x} = -2$ was more convincing in this part of the question than was the rejection of $e^k = -1$ in part (a)(ii). Part (b)(ii) was usually correct.